

Competitive Programming

Section 19: Dijkstra And Weighted Modeling

Competitive Programming Course

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1 Why Dijkstra

BFS solves shortest paths when every move costs the same. Once edges have different nonnegative costs, a path with fewer edges may be more expensive:

```
1 -> 2 cost 100
1 -> 3 cost 1
3 -> 2 cost 1
```

BFS would see vertex 2 in one edge. The cheapest route is actually 1 -> 3 -> 2 with cost 2.

Dijkstra's algorithm is the standard tool for shortest paths with nonnegative edge weights.

2 Recognition

Statement signal	Likely tool
shortest path with nonnegative weights	Dijkstra
grid with tolls/costs, not just steps	Dijkstra on cells
one-time option changes cost	Dijkstra on expanded states
all weights are 0 or 1	next section: 0-1 BFS
negative edge weights	not Dijkstra

2.1 Nonnegative is a requirement, not a hint

Before writing a heap, inspect **every transition cost in the graph you are actually going to run on**. Dijkstra requires all of them to be at least zero. That includes costs introduced by a grid model or a state expansion, not merely the numbers visible in the original statement.

For example, suppose the source has edges $s \rightarrow a$ of cost 1 and $s \rightarrow b$ of cost 2, while $b \rightarrow a$ has cost -5. A heap pops and finalizes a at cost 1 before it pops b . But the route $s \rightarrow b \rightarrow a$ costs -3. The finalization claim is false, so ordinary Dijkstra is invalid.

Use Bellman-Ford or a problem-specific method when negative edges are allowed. If negative cycles are relevant, the question may not even have a finite shortest-path answer. Do not try to repair this by changing the visited array: the theorem, not the implementation detail, has failed.

3 Priority queue primer

Dijkstra repeatedly expands the not-yet-final vertex with smallest known distance. A min-priority queue gives that vertex efficiently.

C++ min-heap:

```
priority_queue<pair<long long,int>,  
              vector<pair<long long,int>>,  
              greater<pair<long long,int>>> pq;
```

Python:

```
import heapq  
heapq.heappush(heap, (distance, vertex))  
distance, vertex = heapq.heappop(heap)
```

4 Dijkstra template

```
dist[start] = 0
push (0, start)

while heap is not empty:
    d, u = pop smallest
    if d is not dist[u]:
        continue
    for each edge u -> v with weight w:
        if d + w < dist[v]:
            dist[v] = d + w
            push (dist[v], v)
```

The `if d is not dist[u]` line skips stale heap entries. We do not decrease a key inside the heap; we push a new better pair and ignore the old one later.

5 The invariant

When Dijkstra pops a non-stale pair $(\text{dist}[u], u)$, $\text{dist}[u]$ is final: no future relaxation can make it smaller.

Why nonnegative weights matter (and why the previous counterexample does not contradict it):

Every future path must first leave the heap through a distance $\geq \text{dist}[u]$.
Adding nonnegative edges cannot reduce that distance below $\text{dist}[u]$.

With a negative edge, this reasoning breaks.

6 Worked example

Graph:

```
1 --4--> 2 --7--> 4
1 --2--> 3 --5--> 4
3 --1--> 2
```

Trace:

```
dist[1]=0
pop 1: set dist[2]=4, dist[3]=2
pop 3: improve dist[2]=3, set dist[4]=7
pop 2: no improvement to 4 because 3+7=10
pop stale (4,2): skip
pop 4: final 7
```

7 Grid Dijkstra

A weighted grid is a graph:

```
cell = vertex  
moving into a neighbor adds that neighbor's toll
```

For a cell (r, c) , relax its four neighbors:

```
new distance = dist[r][c] + toll[nr][nc]
```

If the start cell's toll counts, initialize:

```
dist[0][0] = toll[0][0]
```

8 State expansion

Sometimes a vertex alone is not enough state. In Discount Route, the important state is:

```
(vertex, discount_used)
```

There are two layers:

```
layer 0: discount not used yet  
layer 1: discount already used
```

For an edge $u \rightarrow v$ with cost w :

```
(u,0) -> (v,0) cost w  
(u,0) -> (v,1) cost floor(w/2)  
(u,1) -> (v,1) cost w
```

Then run ordinary Dijkstra on this expanded graph.

9 Complexity

For adjacency-list graphs:

```
Time:   $O((n + m) \log n)$   
Memory:  $O(n + m)$ 
```

For two-layer state graphs, the vertex and edge counts are only multiplied by a constant, so the asymptotic complexity stays the same.

10 Implementation notes

Use 64-bit integers for distances in C++:

```
const long long INF = (1LL << 62);
```

In Python, use a large integer sentinel:

```
inf = 10**30
```

Do not mark a vertex visited when it is pushed. Dijkstra may find a better distance before it is popped. The safe moment is when a non-stale pair is popped.

11 Local exercises

11.1 A. Weighted Route Queries

Run Dijkstra once from the source. Answer each target query from `dist`.

11.2 B. Grid Toll Path

Run Dijkstra over grid cells. The cost includes the start cell and every entered cell.

11.3 C. Discount Route

Run Dijkstra over `(vertex, used)` states. The answer is:

```
min(dist[n][0], dist[n][1])
```

12 Submission checklist

```
Are all edge weights nonnegative?  
Did I use a min-heap?  
Did I skip stale heap entries?  
Did I use long long for distances in C++?  
For grids, did I define whether the start cell cost counts?  
For state graphs, did I include every legal transition?
```

13 Exercises

Solve A with duplicate heap entries, B by turning cells into weighted graph states, and C by adding the used-discount bit to the state. Test stale heap entries, an unreachable destination, and a discount used on the first or last edge.